ESE 101:

Homework 1 (due October 16, 2025):

Show all work to receive full credit. All final answers should have the correct units.

1. *Low-e glazing*. When we were remodeling Linde Laboratory, we wanted it to become energy efficient but were constrained by the mandate to preserve its historical appearance. For example, we could not change the appearance of the windows. However, the window glazing in the historical windows is new.

    (a) We decided double glazing (two glass panes with an air gap in between) to reduce heat conduction was not going to change the energy efficiency of the building much. Can you think of reasons why? (Strong hint: look at the windows and think about what the frames are made of.)

    (b) In your own words, define albedo and emissivity, and explain their relationship. Be precise in your language!

    (c) Regular glass has an emissivity of about 0.9 for thermal infrared radiation. If the outside temperature is 310 K, what is the energy flux with which the windows radiate into the building? What is the wavelength of maximum emission?

    (d) Using glass with a low-emissivity ("low-e") metal oxide coating was found to lead to substantial energy savings. The coating reduces the thermal infrared emissivity of the glass, to a value around 0.04. Qualitatively describe why this reduction in emissivity reduces the building's energy demand.

2. Calculate the root-mean-square (rms) velocity of Nitrogen gas and Oxygen gas at 300K, 275K, and 250K.

3. See coding exercise